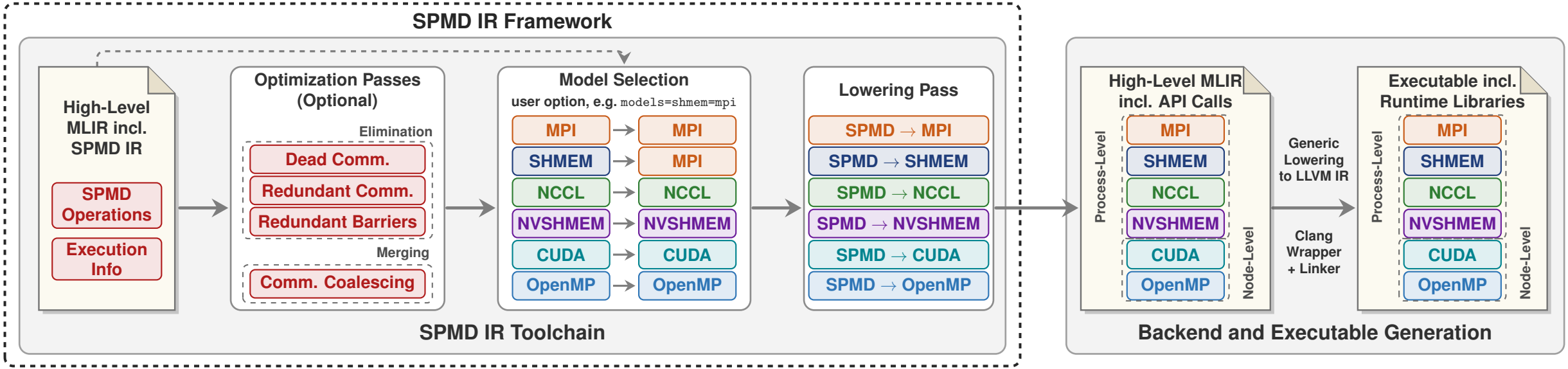
SPMD IR Framework
High-Level MLIR incl. SPMD IR
SPMD Operations
Execution Info
Optimization Passes (Optional)
Elimination
Dead Comm.
Redundant Comm.
Redundant Barriers
Merging
Comm. Coalescing
Model Selection
user option, e.g. models=shmem=mpi
MPI → MPI
SHMEM → MPI
NCCL → NCCL
NVSHMEM → NVSHMEM
CUDA → CUDA
OpenMP → OpenMP
Lowering Pass
SPMD → MPI
SPMD → SHMEM
SPMD → NCCL
SPMD → NVSHMEM
SPMD → CUDA
SPMD → OpenMP
SPMD IR Toolchain
High-Level MLIR incl. API Calls
Process-Level
MPI
SHMEM
NCCL
NVSHMEM
CUDA
OpenMP
Node-Level
Generic Lowering to LLVM IR
Clang Wrapper + Linker
Executable incl. Runtime Libraries
Process-Level
MPI
SHMEM
NCCL
NVSHMEM
CUDA
OpenMP
Node-Level
Backend and Executable Generation